

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Design Task

Code of Conduct:

*I **NARAPPAGARI SRAVYA(192472192)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:SRAVYA

ANNEXURE A

DESIGN TASK

1. Hybrid AI Recommendation Engine Design

Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.

2. Smart Disaster Detection System

Develop a distributed AI system using PySpark and RDD operations to detect natural disasters (such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.

3. AI-Based Fraud Detection Framework

Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.

4. Autonomous Supply Chain Optimization System

Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory, logistics, demand forecasting). Describe how RDD transformations support largescale data processing and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.

5. Personalized Learning Analytics Platform

Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

RUBRICS FOR CALCULATION

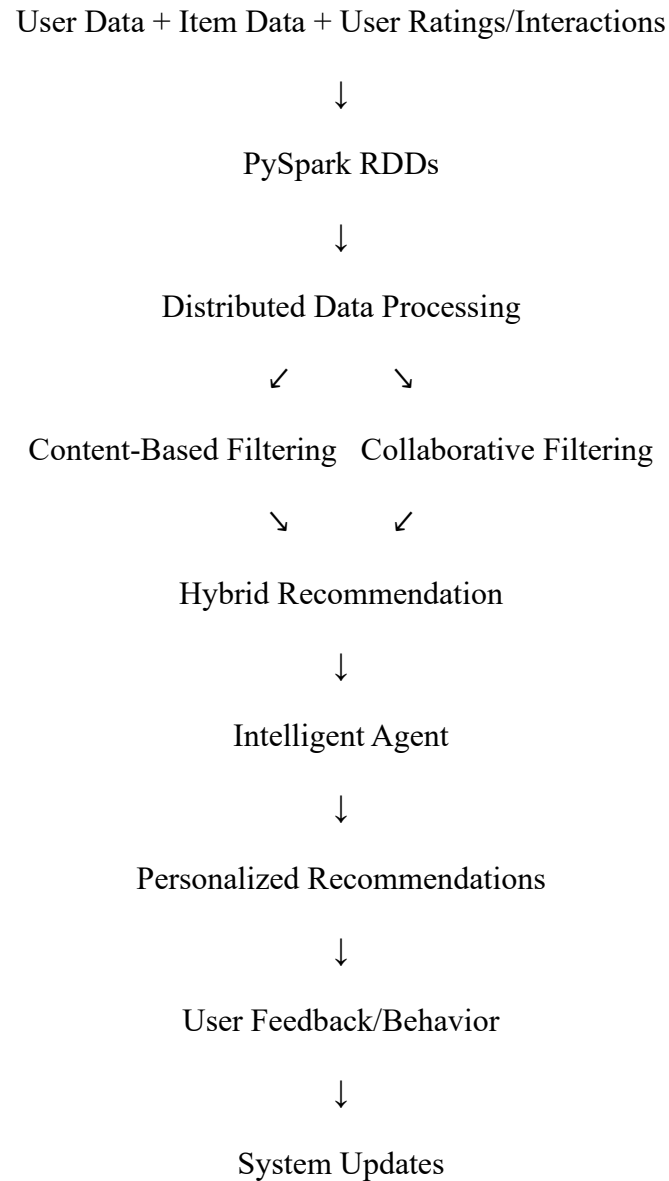
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

1. Hybrid AI Recommendation Engine Design

Objective

To design a recommendation system that combines **Content-Based Filtering** and **Collaborative Filtering** using **PySpark RDDs**.

System Architecture



Working

1. Data Collection

The system collects:

- User preferences
- Product or item information

- Ratings
- Clicks and searches
- Purchase history

2. Content-Based Filtering

Items are recommended based on their features.

Example:

If a user likes AI and Machine Learning books, the system recommends similar books.

3. Collaborative Filtering

Recommendations are generated using the behavior of similar users.

Example:

If two users have similar interests, items liked by one user can be recommended to the other.

4. Hybrid Recommendation

Both techniques are combined to produce better recommendations.

Role of PySpark RDDs

RDD operations process large datasets across multiple computers.

Transformations:

- `map()` – processes user-item data
- `filter()` – removes irrelevant interactions
- `reduceByKey()` – combines ratings
- `join()` – combines user and item information

Actions:

- `collect()`
- `count()`
- `reduce()`

Role of Intelligent Agents

Intelligent agents:

- Monitor user behavior

- Analyze clicks and ratings
- Learn user preferences
- Update recommendations in real time
- Use feedback to improve future recommendations

Conclusion

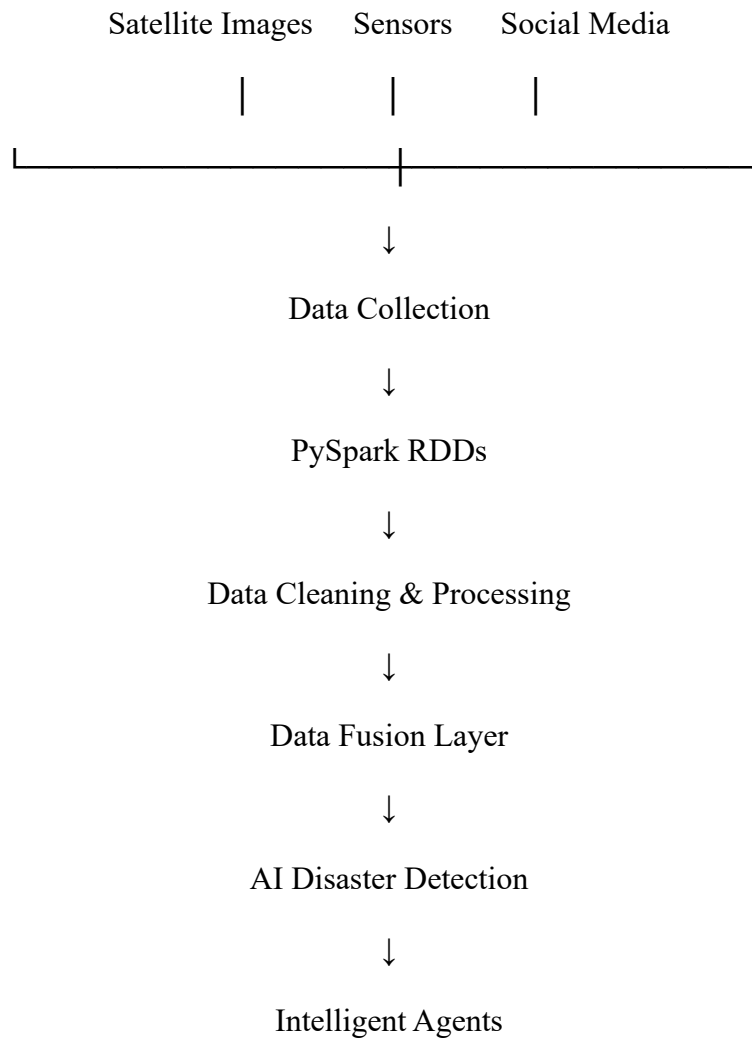
The hybrid system provides **more accurate and personalized recommendations** while PySpark allows the system to handle millions of user-item interactions efficiently.

2. Smart Disaster Detection System

Objective

To detect natural disasters such as **floods and earthquakes** using distributed AI and multiple data sources.

System Architecture





Warning and Response System

Data Sources

The system collects information from:

- **Satellite images** – detect floods, fires, and damaged areas
- **Sensors** – measure water level, temperature, and seismic activity
- **Social media** – identify emergency reports and disaster-related messages

Data Fusion

Data fusion means combining information from different sources.

For example:

- Satellite image shows flooding
- Water sensor shows high water level
- Social media reports flooding

When all sources provide similar information, the system can identify the disaster with higher confidence.

Role of PySpark RDDs

PySpark distributes large amounts of disaster data across multiple machines.

Common operations include:

- `map()` – process individual sensor records
- `filter()` – identify abnormal values
- `union()` – combine multiple data sources
- `reduceByKey()` – summarize information by location

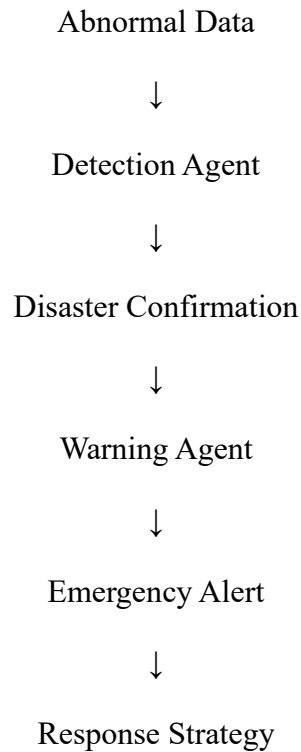
Intelligent Agents

Different agents perform different tasks:

- **Monitoring Agent** – monitors incoming data
- **Detection Agent** – identifies disasters
- **Warning Agent** – sends alerts

- **Response Agent** – suggests emergency actions

Early Warning Process



Conclusion

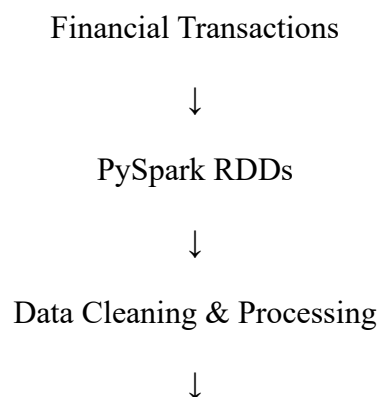
The system uses distributed processing and intelligent agents to provide **fast disaster detection, early warnings, and coordinated emergency responses.**

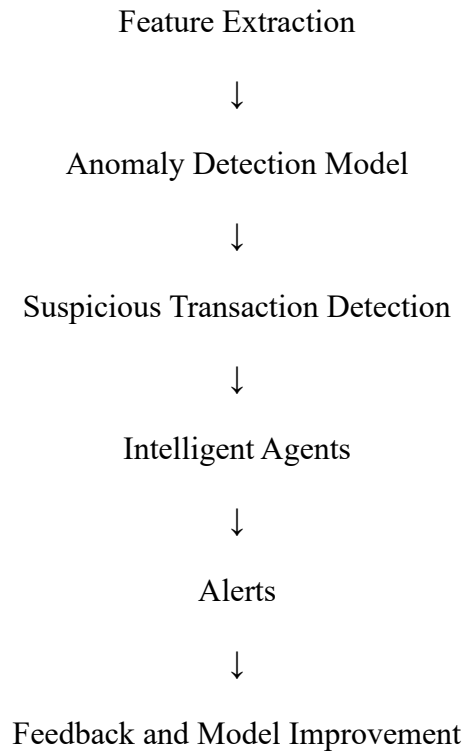
3. AI-Based Fraud Detection Framework

Objective

To design a scalable fraud detection system for analyzing large volumes of financial transactions.

System Architecture





Data Processing

The system analyzes:

- Transaction amount
- Transaction location
- Transaction time
- User spending behavior
- Device information

Anomaly Detection

Fraud can be detected by identifying unusual behavior.

Examples:

- Very large transaction amount
- Multiple transactions in a short time
- Transaction from an unusual location
- Unusual spending pattern

PySpark RDD Operations

- `map()` – extract transaction features
- `filter()` – select suspicious transactions
- `reduceByKey()` – analyze transactions for each user
- `groupByKey()` – group transactions

Because RDDs are distributed, millions of transactions can be processed in parallel.

Intelligent Agents

Monitoring Agent

Continuously monitors transactions.

Detection Agent

Uses anomaly detection to identify suspicious activities.

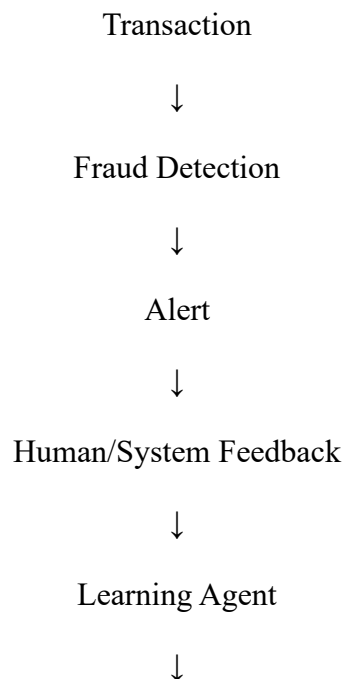
Alert Agent

Sends alerts when fraud is detected.

Learning Agent

Uses feedback from confirmed fraud cases to improve the system.

Continuous Improvement



Conclusion

The framework provides **scalable, fast, and continuously improving fraud detection** using PySpark and intelligent agents.

4. Autonomous Supply Chain Optimization System

Objective

To optimize supply chain activities such as:

- Inventory management
- Transportation
- Logistics
- Demand forecasting

System Architecture

Sales Data + Inventory Data + Logistics Data



PySpark RDDs



Distributed Data Processing



AI Analytics



Inventory Demand Logistics

Agent Agent Agent



Decision System



Optimized Supply Chain

Role of RDD Transformations

RDD transformations help process large supply chain data.

map()

Processes individual records.

filter()

Removes unnecessary or invalid data.

reduceByKey()

Calculates totals such as product demand.

join()

Combines inventory and sales information.

groupByKey()

Groups data based on location or product.

Intelligent Agents

Inventory Agent

Monitors stock levels and decides when to reorder products.

Demand Forecasting Agent

Predicts future product demand.

Logistics Agent

Selects efficient transportation routes.

Coordination Agent

Coordinates decisions between different agents.

Example

If demand for a product increases:

1. Demand Agent detects increased demand.
2. Inventory Agent checks stock.
3. Logistics Agent plans transportation.

4. The system automatically updates the supply chain plan.

Benefits

- Reduces operational cost
- Prevents overstocking
- Prevents stock shortages
- Improves delivery speed
- Increases efficiency

Conclusion

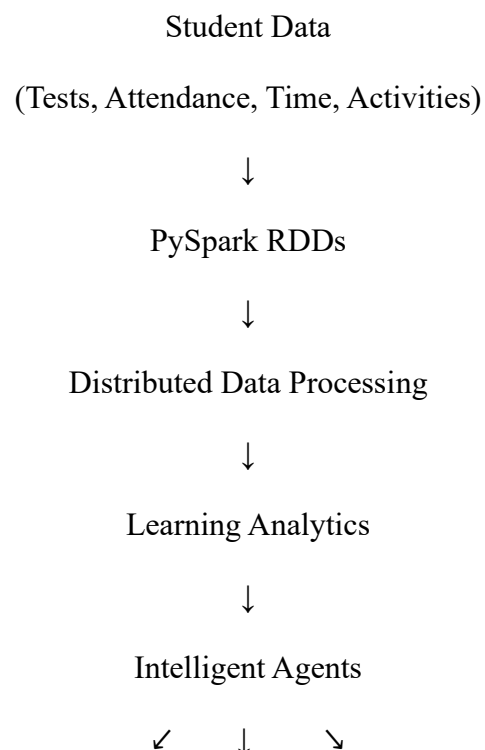
The autonomous supply chain system uses **distributed data processing and multiple intelligent agents** to make fast and decentralized decisions.

5. Personalized Learning Analytics Platform

Objective

To create an AI-based educational platform that analyzes student learning behavior and provides personalized learning.

System Architecture



Content Performance Feedback

Agent Agent Agent



Personalized Learning



Student Feedback



System Update

Data Collected

The platform analyzes:

- Test marks
- Assignment performance
- Attendance
- Time spent learning
- Learning activities
- Quiz results

Role of PySpark RDDs

PySpark processes data from thousands of students simultaneously.

RDD operations:

- `map()` – process student records
- `filter()` – identify weak students
- `reduceByKey()` – calculate average performance
- `groupByKey()` – group students by performance level

Intelligent Agents

Student Performance Agent

Analyzes student marks and performance.

Content Recommendation Agent

Suggests appropriate learning materials.

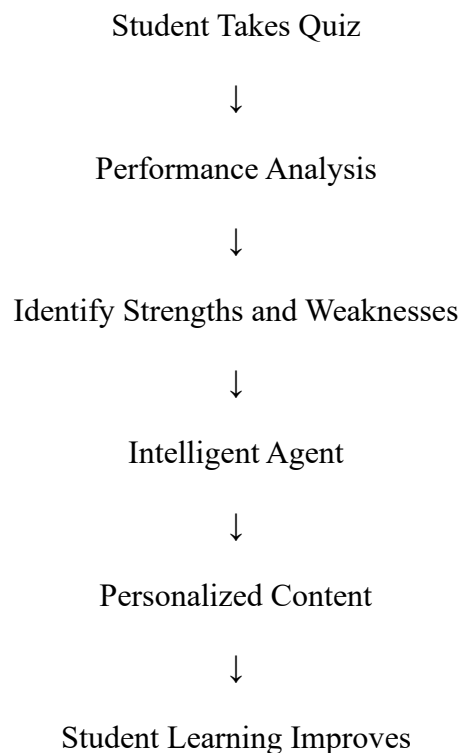
Feedback Agent

Provides immediate feedback.

Adaptive Learning Agent

Changes the difficulty level based on student performance.

Adaptive Learning Example



Real-Time Feedback

The system can provide:

- Immediate quiz feedback
- Recommended videos or notes
- Additional practice questions
- Easier or advanced content based on performance

Scalability

Using PySpark, the platform can process data for **thousands or millions of learners** by distributing the workload across multiple computers.

Conclusion

The Personalized Learning Analytics Platform combines **PySpark distributed analytics, AI, and intelligent agents** to provide scalable, adaptive, and personalized education.